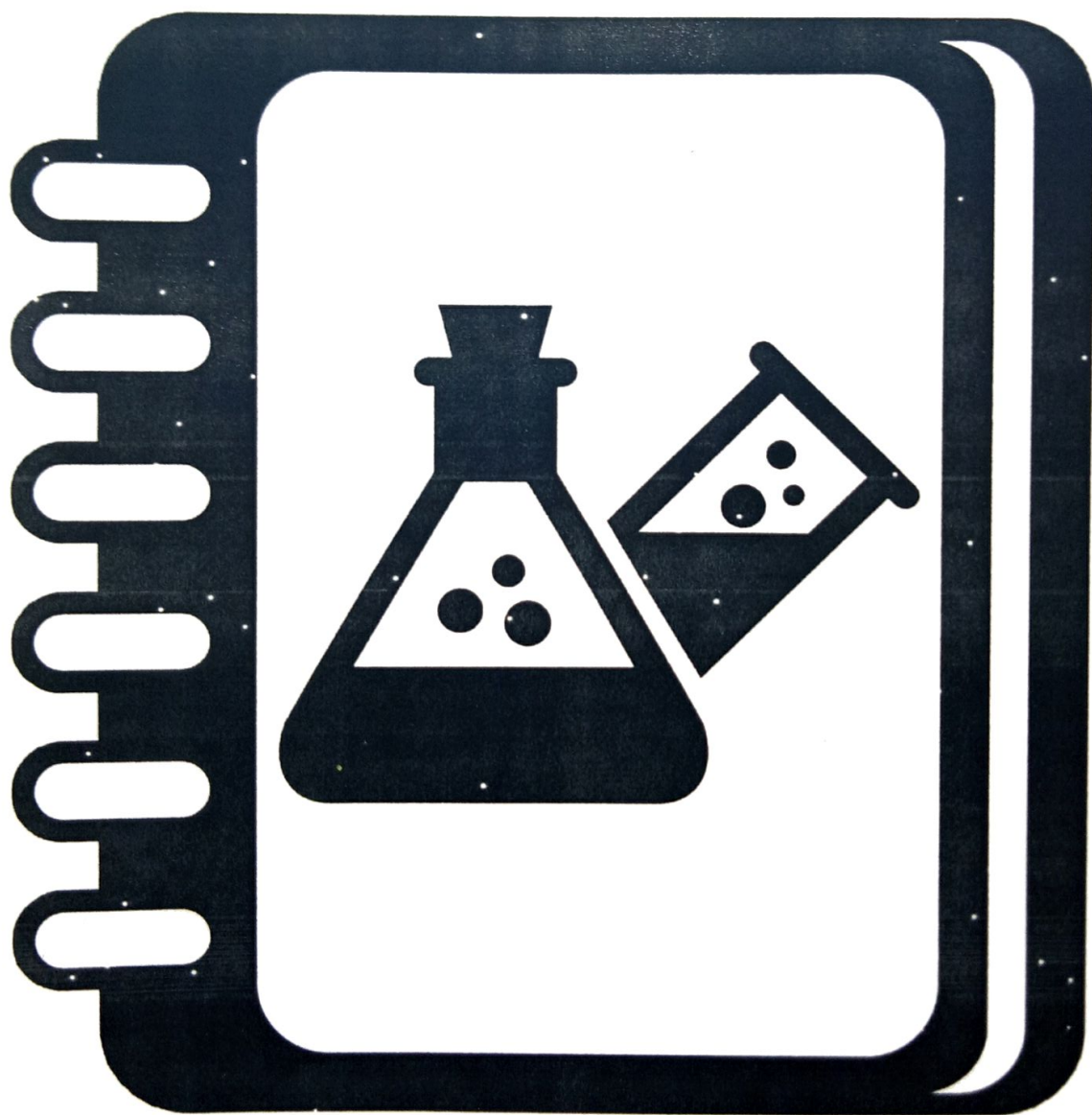




UPLINK

TUTORIAL AND REVIEW CENTER



CHEMISTRY

LESSON 1: MATTER

1.1 STATES OF MATTER

CHEMISTRY is the study of matter – its composition, properties and the changes it undergoes.

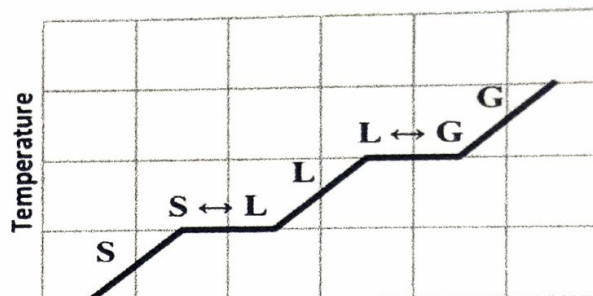
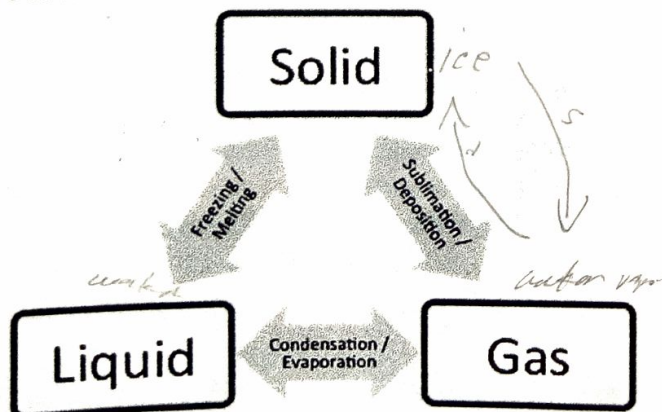
MATTER is anything that has mass and occupies space.

MASS is the amount of matter contained in an object.

VOLUME is the amount of space occupied by matter.

STATES OF MATTER	
SOLID	
- compact	
- definite shape and volume	HEAT
- incompressible	
- usually have low temperature	temp phase
LIQUID	
- compact to slightly spaced	
- indefinite shape but definite volume	
- incompressible	
- usually have low to high temperature	
GAS	
- spaced apart	
- indefinite shape and indefinite volume	
- compressible	
- usually have high temperature	

PHASE CHANGES



1.2 PROPERTIES OF MATTER

INTENSIVE PROPERTIES

- properties that does NOT depend on the amount of matter in a substance

Boiling, Freezing, Melting Points	conditions at which matter changes from one phase to another
Density	mass per unit volume $P_{\text{mass}} = m/v$
Buoyancy	ability to float
Ductility	ability to be drawn out unto wires
Malleability	ability to be hammered into sheets
Elasticity	ability to return to its original shape after being deformed
Viscosity	ability to resist flow

EXTENSIVE PROPERTIES

- properties that depend on the amount of matter in a substance.

Mass	amount of matter present in a substance
Volume	space occupied by an object
Energy	amount of energy stored in an object
Entropy	measure of disorder of a thermodynamic system
Electrical Charge	the total charge of the particles of an object

DENSITY

- mass per unit volume

$$\rho = \frac{m}{V}$$

SPECIFIC GRAVITY

- ratio of the density of a sample and the density of a reference substance (usually water).

$$SG = \frac{\rho_{\text{sample}}}{\rho_{\text{reference}}}$$

APPARENT SPECIFIC GRAVITY

- ratio between the weights of equal volumes of a sample and water.

$$SG_{\text{apparent}} = \frac{W_{\text{sample}}}{W_{\text{water}}}$$

TRUE SPECIFIC GRAVITY

- ratio between the weights in a vacuum of equal volumes of a sample and water.

$$SG_{\text{true}} = \frac{W_{V_{\text{sample}}}}{W_{V_{\text{water}}}}$$

Specific gravities can be used to determine whether a sample would float on another substance. If the SG of that sample is less than 1, it will float. For example, the SG of helium in air is 0.139.

Practice Problem 1:

Particles of this phase of matter are strongly bound to each other, making them immobile. What phase of matter is the statement referring to?

- Solid
- Liquid
- Gas

1.3 CLASSIFICATION OF MATTER**1.3.1 PURE SUBSTANCES**

- made of only one kind of substance
 - ELEMENTS** – made of only one kind of atom.
 - Non-Metals** – the number of valence electrons is close to completing the octet rule; accepts electrons during ionic bonding; forms **anions** when ionized
Example: *Oxygen*
 - Metals** – the number of valence electrons is small; donates electrons during ionic bonding; forms **cations** when ionized
Example: *Potassium*
 - Metalloids** – can act as a metal or a non-metal depending on the conditions
Example: *Silicon*
 - COMPOUNDS** – made up of only one type of molecule (two or more atoms that are chemically bonded)
 - Ionic** – components are a pair of anion group and a cation group; electrons are transferred from the cation to the anion.
Example: *NaCl*
 - Covalent** – components are both anion groups; electrons are shared within the bond.
Example: *CO₂*
 - Metallic** – components are both metals; electrons are stored in a “sea of electrons”.
Example: *AgNO₃*

1.3.2 MIXTURES

unsaturated → made up of two or more different kinds of substances

- i. **HOMOGENOUS** – mixtures wherein only one phase is discernible; commonly called “solutions”

CONCENTRATION – measure of the relative amounts of the components of a solution

SOLUTE – the component that dissolves in the

Molarity (M)	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
Molality (m)	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mass Percent	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 100$
Volume Percent	$\frac{\text{vol of solute}}{\text{vol of solution}} \times 100$
Mole Fraction	$\frac{\text{moles of component}}{\text{total moles}}$

solution

SOLVENT – the component that dissolves other components of the solution

SOLUBILITY – the maximum amount of solute that can be dissolved in a particular solvent under specific conditions; often expressed as a ratio or other measures of concentration

SOME COMMON MEASURES OF CONCENTRATION

Solubility greatly depends on the balance of the intermolecular forces of the solute and the solvent.

SOLUBILITY RULE

- All nitrates (NO_3^-) are soluble
- All chlorides (Cl^-) are soluble except AgCl , Hg_2Cl_2 , and PbCl_2
- Most sulfates (SO_4^{2-}) are soluble; exceptions include BaSO_4 , PbSO_4 , and SrSO_4
- All carbonates (CO_3^{2-}) are insoluble except NH_4^+ and those of Group 1
- All hydroxides (OH^-) are insoluble except those of Group 1, $\text{Ba}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, and $\text{Ca}(\text{OH})_2$
- All sulfides are insoluble except those of Group 1, Group 2 and NH_4^+

- ii. **HETEROGENOUS** – mixtures wherein two or more distinct phases are discernible

a. SUSPENSIONS

- particles are too large and too heavy to be carried by the particles they are suspended in, causing them to settle after some time

b. COLLOIDS

- suspended particles are small and light

TYPES OF COLLOIDS		
PHASE	NAME	EXAMPLE
Solid in Solid	Solid Sol	colored glass, ruby, opal
Solid in Liquid	Sol	paint, blood
Solid in Gas	Solid Aerosol	smoke, ice clouds, air particulates
Liquid in Solid	Gel	agar, gelatin
Liquid in Liquid	Emulsion	egg, milk, hand cream
Liquid in Gas	Aerosol	fog, cloud, hair spray
Gas in Solid	Solid Foam	Styrofoam, aerogel, pumice
Gas in Liquid	Foam	shaving cream, whipped cream

enough to be carried by the movement of the particles in which they are suspended in.

Brownian Motion – movement of the particles in a substance

Tyndall Effect – scattering of light as it passes through a colloid

LESSON 2: ATOMS

2.1 ATOMS

ATOMIC THEORY OF MATTER

- All matter is made of atoms
- Atoms of the same element are identical
- Atoms can combine to produce other substances
- Atoms combine in specific proportion to form a specific substance

COMPONENTS OF AN ATOM

- Protons (p^+)** – positively charged sub-atomic particle with a mass of 1.67×10^{-27} kg or 1 atomic mass unit (amu)
- Neutrons (n^0)** – neutral in charge and has a mass of 1.67×10^{-27} kg or 1 amu
- Electrons (e^-)** – negatively charged particles with a mass of 9.11×10^{-31} kg that is considered to be negligible

Protons and **neutrons** are found within the center of the atom called “**nucleus**” and is where the mass of the whole atom is concentrated.

Electrons are located orbiting **outside the nucleus** in certain regions depending on their quantum state

2.2 IONS, ISOTOPES, AND ISOMERS

ATOMIC NUMBER (Z)

- equal to the number of protons

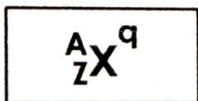
MASS NUMBER (A)

- equal to the sum of the number of protons and neutrons

CHARGE (q)

- equal to the difference between number of protons and the number of electrons

COMMON NOTATION



IONS – atoms or molecules are charged due to the unequal number of electrons and protons.

Cation – positive ion

Anion – negative ion

ISOTOPES – atoms of the same element that have the same number of protons but different number of neutrons, thus different mass numbers.

ISOMERS – different compounds have the same formula but their atoms are arranged differently.

PERCENT NATURAL ABUNDANCE – relative abundance of a certain isotope compared to all the other isotopes of the same element.

ATOMIC WEIGHT – weighted average of the mass numbers of the isotopes of that particular element.

Practice Problem 2:

An unknown atom contains 12 protons, 12 neutrons and 11 electrons. What are the atom's atomic number and charge?

- 12 and +1
- 24 and +1
- 12 and -1
- 24 and -1

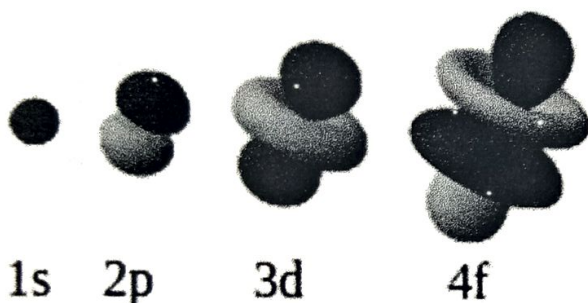
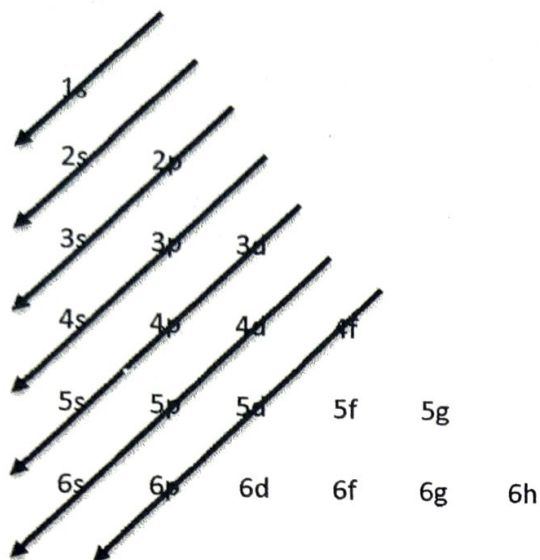
2.3 ELECTRON CONFIGURATION

Energy Level – region wherein electrons are most likely found; also referred to as orbitals

Subsheli – minor energy levels occupied by electrons within the energy level

Categories:

- **Sharp (s):** 2 electrons maximum
- **Principal (p):** 6 electrons maximum
- **Diffuse (d):** 10 electrons maximum
- **Fundamental (f):** 14 electrons maximum

**Quantum Numbers:****Principal Quantum Number (n)**

$$n=1,2,3,\dots$$

Azimuthal Quantum Number (ℓ)

$$\ell = n - 1 \text{ to } 0 \quad \ell=1; \ell=0$$

Magnetic Quantum Number (m_ℓ)

$$m_\ell = -\ell \text{ to } +\ell \quad m_\ell = -1, 0, +1$$

Magnetic Electron Spin (m_s)

$$m_s = -\frac{1}{2} \text{ or } +\frac{1}{2}$$

Rules in Assigning Electron Orbitals:

1. Electrons occupy orbitals such that it minimizes the energy of the atom.
2. **Pauli Exclusion Principle:** No two electrons may have all four quantum numbers alike.
3. **Hund's Rule:** Electrons found in the same energy level will initially occupy these energy levels singly before any pairing of electrons occurs.

Aufbau Process: Electrons are assigned in order of increasing atomic number with levels closest to the nucleus occupied first before the higher energy levels

LESSON 3: PERIODIC TABLE OF ELEMENTS

3.1 CLASSIFICATION OF ELEMENTS

i. METALS

- usually solid, malleable, ductile, and can be plastically deformed
- highly conductive for heat and electricity with the exception of a few elements
- low electronegativities and ionization energies
- high melting points and densities

ii. NON-METALS

- brittle solids
- poor conductors of heat and electricity
- high electronegativities and ionization energies

iii. METALLOIDS

- properties are in between those of metals and non-metals
- reactivity depends on the elements to be reacted with

3.2 PERIODS AND GROUPS OF ELEMENTS

i. GROUP A

a. Grp 1A (Alkali Metals)

- Soft metallic solids; shiny and lustrous; reacts vigorously with water to produce hydroxides

b. Grp 2A (Alkaline Earth Metals)

c. Grp 3A (Boron Grp or Earth Metals)

b. Grp 4A (Carbon Grp or Tetrels)

c. Grp 5A (Nitrogen Grp or Pnictogens)

d. Grp 6A (Oxygen Grp or Chalcogens)

e. Grp 7A (Halogens)

f. Grp 8A (Noble Gases)

- Harder than 1A metals; shiny, lustrous and oxidizes easily
- Reactive non-metals
- Unreactive under ordinary circumstances; usually found as monatomic gases

ii. GROUP B

a. Transition Metals

- Hard metallic; shiny and lustrous; dense and have high melting points; usually have high oxidation states

b. Lanthanide Series

c. Actinide Series

3.3 PERIODIC TRENDS

i. Electro Negativity

- the tendency of an electron to bond
- increases from left to right along a period
- decreases from top to bottom along a column/group

ii. Ionization Energy

- energy needed to ionize an atom
- increases from left to right along a period
- decreases from top to bottom along a column/group

iii. Electron Affinity

- energy released when an electron is added to an atom
- positive when energy is released, negative when absorbed
- generally increases from left to right along a period
- generally decreases from top to bottom along a column/group

iv. Atomic Size/Radius

- the farthest extent of the presence of electrons
- decreases from left to right along a period
- increases from top to bottom along a column/group

Practice Problem 3:

Which of the following atoms would you expect to have the largest atomic radius?

- Lithium
- Calcium
- Potassium
- Bromine

LESSON 4: COMPOUNDS

Compounds are formed when two or more atoms of different elements chemically combine.

4.1 CHEMICAL BONDING

TYPES OF COMPOUNDS AND CHEMICAL BONDS

i. IONIC

- form crystals
- high melting and boiling points
- hard and brittle
- poor conductors when solid, good conductors when dissolved in water

ii. COVALENT

- most have relatively low melting and boiling points
- tend to be soft and relatively flexible
- tend to be more flammable
- many are insoluble in water and those that do dissolve do not conduct electricity

iii. METALLIC

- have moderately high melting points
- malleable and ductile
- good conductors of heat and electricity

4.1.1 MOLECULAR GEOMETRY

INTERMOLECULAR FORCES

i. Electrostatic Interactions

- attraction between the opposing poles of dipole molecules
 - Dipole-Dipole
 - Ion-Dipole
 - Ion-Induced Dipole
 - Hydrogen Bonding

ii. Van Der Waal's

- sum of all the attractive and repulsive forces between and within molecules

EFFECTS OF INTERMOLECULAR FORCES

- Intermolecular forces are major factors that determine some properties of substances.

SOME PROPERTIES AFFECTED BY IMFA

Crystalline Formation	occurs when IMFA creates a regular structural pattern
Vapor Pressure	fluid motion create an internal pressure that is limited by the strength of the IMFA
Boiling and Melting Point	condition at which boiling occurs depend on the vapor pressure
Surface Tension	ability of liquids to form a shape with the smallest surface area
Capillarity	tendency of liquids to rise or fall within a narrow tube
Adhesion / Cohesion	attraction between like or unlike substances
Viscosity	a liquid's resistance to flow
Diffusion	spreading out of fluid particles to occupy available space
Heat Capacity	heat required to raise the temperature of 1 g of a substance by 1°C
Critical Point	conditions a which the liquid and gaseous phases of a substance become indistinguishable

4.2 CHEMICAL FORMULA AND NOMENCLATURE

4.2.1 IONIC COMPOUNDS

The name of ionic compounds is determined by the names of its components. Cations are named first before anions. However, the anion will end with *-ide*.

Ex. NaCl – *Sodium chloride*

4.2.2 COVALENT COMPOUNDS

Numeral prefixes are used in naming covalent compounds.

Ex. CO₂ – *Carbon dioxide*

4.2.3 POLYATOMIC IONS

These are group of covalently bonded atoms that carries an electric charge.

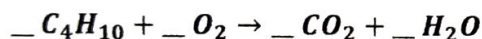
Ex. Na MnO₄⁻ - *Sodium permanganate*

COMMON POLYATOMIC IONS

Ammonium	NH_4^+
Acetate	$\text{C}_2\text{H}_3\text{O}_2^-$
Carbonate	CO_3^{2-}
Chromate	CrO_4^{2-}
Cyanide	CN^-
Hydroxide	OH^-
Permanganate	MnO_4^-
Peroxide	O_2^{2-}
Phosphate	PO_4^{3-}
Sulfate	SO_4^{2-}
Sulfite	SO_3^-
Nitrate	NO_3^-

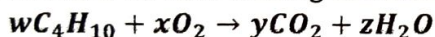
LESSON 5: CHEMICAL REACTIONS

5.1 BALANCING CHEMICAL EQUATIONS



ALGEBRAIC METHOD

- Assign variables for each missing coefficient



- Create equations to balance each element

$$\text{C: } 4w = y \quad \text{H: } 10w = 2z$$

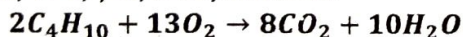
$$\text{O: } 2x = 2y + z$$

- Choose any variable and assign it a value of 1 and solve for the values of the others

$$\text{If } w=1, \text{ then } y=4, z=5, \text{ and } x=13/2$$

- Eliminate all the fractions by multiplying everything by the LCD

$$\text{So, } w=2, y=8, z=10, \text{ and } x=13$$



5.2 TYPES OF CHEMICAL REACTIONS

Synthesis	$A + B \rightarrow AB$
Analysis	$AB \rightarrow A + B$
Replacement	$AB + C \rightarrow AC + B$
Double Displacement	$AB + CD \rightarrow AD + CB$

SOME SPECIAL TYPES:

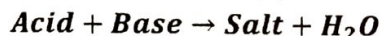
i. Combustion



ii. Redox Reaction

-electron transfer from one atom to another

iii. Neutralization Reaction



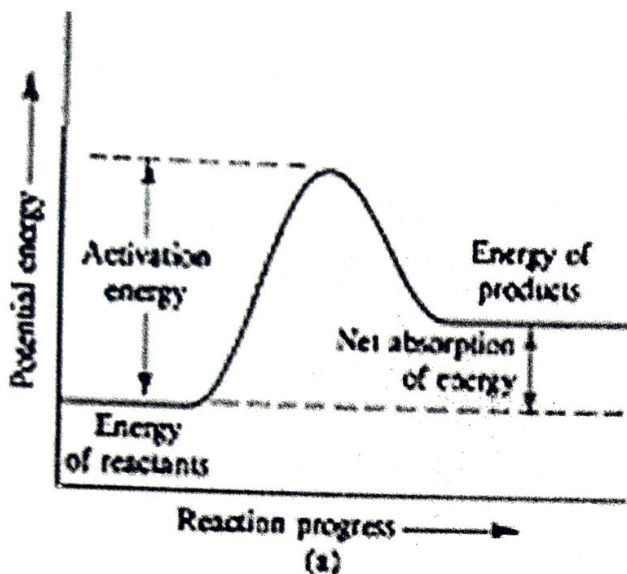
5.3 ENDOTHERMIC VS. EXOTHERMIC REACTION

ENERGIES, SPONTANEITY, EQUILIBRIUM

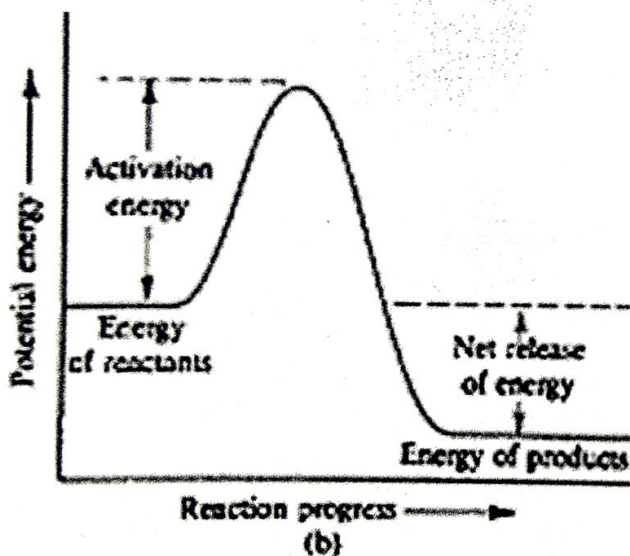
Any object contains a specific amount of energy. In a chemical reaction, the energies of the reactant change into that of the product.

ENTHALPY - the total energy of an object or system

ENDOTHERMIC - reactions are those that require addition of energy. This means that the product has a higher enthalpy than the reactants.



EXOTHERMIC - reactions are those that require removal of energy, which means that the product has a lower enthalpy than the reactant.



ACTIVATION ENERGY - energy required to start a reaction

SPONTANEOUS REACTIONS - reactions that require very small activation energy and can start without external interference

CATALYST - external influence that can initiate or hasten a reaction without tampering with the product

REVERSIBLE REACTIONS - reactions whose products can spontaneously react to form the original reactants thereby reversing the reaction

EQUILIBRIUM - state of reaction at which the rate of the forward reaction is the same as the rate of the reverse reaction producing a condition where the concentrations of the reactants and the products remain relatively constant

LESSON 6: STOICHIOMETRY

6.1 MOLE CONCEPT

THE MOLE - relative amount of particles present in a sample.

THE AVOGADRO'S NUMBER

$$6.022 \times 10^{23}$$

- the amount of particles present in each mole of a substance

CALCULATIONS IN STOICHIOMETRY

Mass to Mass

$$\text{Mass A} \times \frac{1 \text{ mol A}}{\text{MW A}} \times \frac{\text{coef A}}{\text{coef B}} \times \frac{\text{MW B}}{1 \text{ mol B}} = \text{Mass B}$$

Mole to Mole

$$\text{Mole A} \times \frac{\text{coef A}}{\text{coef B}} = \text{Mole B}$$

TIP: Treat the equation as a ratio of the chemical equations and simply calculate the number of moles using "ratio and proportion"

Practice Problem 4:

How many moles of sodium is 1.20×10^{24} atoms of sodium?

6.2 FORMULA OR MOLECULAR MASS

FORMULA MASS is for compounds that are made up of ions and have primarily ionic bonding.

MOLECULAR MASS is for compounds with primarily covalent bonding.

Practice Problem 5:

Calculate the formula or molecular mass of the following compounds:

- $\text{Al}(\text{OH})_3$
- $\text{C}_7\text{H}_5\text{NO}_3\text{S}$
- N_2
- MgCO_3

6.3 EMPIRICAL AND MOLECULAR FORMULA

Empirical Formula shows the simplest ratio of the number of atoms of each component elements

Molecular Formula shows the actual number of each of the atom of the component elements

Structural Formula shows the approximate arrangement of the atoms within the molecule

6.4 PERCENT YIELD

THEORETICAL YIELD is the yield when a given quantity of a reactant is *completely* converted to a product.

ACTUAL YIELD is the actual quantity obtained. Loss of some of the products is observed.

$$\text{PERCENTAGE YIELD} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

6.5 LIMITING AND EXCESS REAGENTS

LIMITING REAGENT – reactant that is first to be totally consumed halting further reactions

EXCESS REAGENT – reactant that is left when the limiting reagent is consumed

THE BANANA SPLIT METHOD

- Divide each given by their corresponding coefficients and molar masses. Whichever has the smallest result is the limiting reagent.
- From the result of the above step, subtract the result of the limiting reagent. Multiply to this result the corresponding coefficients and molar masses to get the amount in excess.

6.6 REDOX REACTIONS

- atoms lose or gain electrons during reactions

LOSS of ELECTRONS is **OXIDATION** (Reducing Agent)

GAIN of ELECTRONS is **REDUCTION** (Oxidizing Agent)

OXIDATION STATES/NUMBERS

- The sum of all oxidation numbers is equal to the charge of the molecule
- Molecules with only 1 type of atom has a **ZERO** oxidation state
- HYDROGEN** has a **1+** oxidation state except on rare instances
- OXYGEN** has a **2-** oxidation state except on rare instances
- HALOGENS** has a **1-** oxidation state
- Members of **1A** has a **1+** oxidation state
- Members of **2A** has a **2+** oxidation state

The total amount of electrons gained must be equal to the total amount lost.

OXIDIZING AGENT – reactant that causes oxidation; usually contains the reduced atoms

REDUCING AGENT – reactant that causes reduction; usually contains the oxidized atoms

Practice Problem 6:

Identify which is being oxidized and reduced in the reaction below:



LESSON 7: GASES

THE IDEAL GAS CONCEPT

An Ideal gas is where all interactions between the molecules are only through collisions that are assumed to be completely elastic.

- gases are considered to have mass but, despite being able to collide, have no volume
- exists in the condition called Standard Temperature and Pressure (STP) of 0°C and 1 atm
- at STP, 1 mole of a gas occupies 22.4 L

IDEAL GAS EQUATION:

$$PV = nRT$$

where P is pressure, V is volume, n is the number of moles, T is temperature, R is the Ideal Gas Constant equivalent to $0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}$

- It is important to note that in Gas Laws, temperatures must be measured in Kelvin.

7.1 GAS LAWS

i. Boyle's Law

- a particular amount of gas at a constant temperature increase in volume when the pressure is decreased.

$$P_1 V_1 = P_2 V_2$$

ii. Charles Law

- a particular amount of gas in a constant pressure increase in volume when the temperature is increased.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

iii. Gay-Lussac's Law

- a particular amount of gas in a constant volume increase in temperature when the pressure is increased.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

iv. Avogadro's Law

- a gas in a constant pressure and temperature increases in volume when the amount of particles is increased

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

v. The Combined Gas Law

$$\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$$

vi. Law of Partial Pressures

- the pressure of a mixture of gases is the sum of the pressures of the individual components.

$$P_T = P_1 + P_2 + P_3 \dots$$

vii. Law of Effusion

- the rate at which a gas diffuses is inversely proportional to the root of its density (or mass).

$$\frac{R_1}{R_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

Practice Problem 7:

What is the temperature of 4 moles of a gas kept in a 2-liter container under a pressure of 1.5 atm? ($R = 0.082 \text{ atm} \cdot \text{L} / \text{mole} \cdot \text{K}$)?

- 9.15 K
- 36.59 K
- 0.984 K
- 915 K

LESSON 8: THERMOCHEMISTRY

THERMAL PROPERTIES OF MATTER

- SPECIFIC HEAT CAPACITY (C)** – amount of heat needed to change the temperature of 1 g of a substance by 1°C.
- HEAT OF VAPORIZATION (ΔH_{VAP})** – amount of heat required to transform 1 g of a substance from liquid to gas or gas to liquid
- HEAT OF FUSION (ΔH_{FUS})** – amount of heat required to transform 1 g of a substance from solid to liquid or liquid to solid

8.1 LAWS OF THERMODYNAMICS

HEAT AND ITS EFFECTS

Changes in temperature

$$Q = mC\Delta T$$

Changes in phase

$$Q = m\Delta H$$

LESSON 9: ACIDS AND BASES

9.1 CLASSIFYING ACIDS AND BASES

- Arrhenius Definitions**
Acids contain H^+ while bases contain OH^-
- Brønsted-Lowry Definitions**
Acids donate protons (H^+), while bases accept protons.
- Lewis Definitions**
Acids accept electron pairs, while bases donate electron pairs.

9.2 PROPERTIES OF ACIDS AND BASES

ACIDS are typically sour in taste. They are also corrosive to most metals

BASES are typically bitter in taste and slippery to the touch.

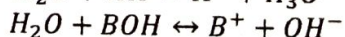
DISSOCIATION OF ACIDS AND BASES

Acids dissociate to produce H^+ ions and anions, while bases dissociate to form OH^- ions and cations.

STRONG ACIDS/BASES - acids/bases that have dissociation constants close to 1.

WEAK ACIDS/BASES - acids/bases that have dissociation constants far from 1.

When acids/bases are in an aqueous state (dissolved in water), they dissociate into their respective ions.



ACIDITY AND BASICITY

Acidity of a substance depends on the concentrations of its dissociated ions.

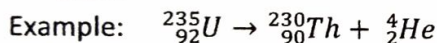
LESSON 10: NUCLEAR CHEMISTRY

- Atomic reactions where changes in the number of nuclear particles change.
- Decrease in the nuclear particles is called **decay**.
- Breaking apart of the nucleus to form new atoms is called **fission**.
- Merging of nuclei to form a new atom is called **fusion**.

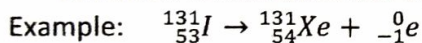
10.1 TYPES OF DECAY

Unstable (radioactive) atoms undergo radioactive decay to become a more stable atom.

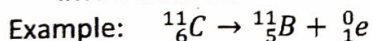
- ALPHA RADIATION** – emits a particle that has 2 protons and 2 neutrons similar to a He atom



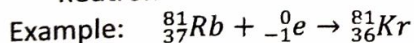
- BETA RADIATION** – emits an electron when a neutron is converted into a proton



- GAMMA RADIATION** – emits photons
- POSITRON EMISSION** – emits a massless positive particle when a proton is converted into a neutron

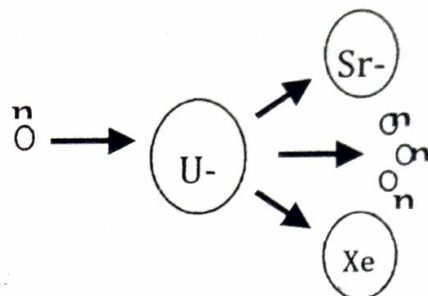


- ELECTRON CAPTURE** – an electron is absorbed and a proton is converted into a neutron

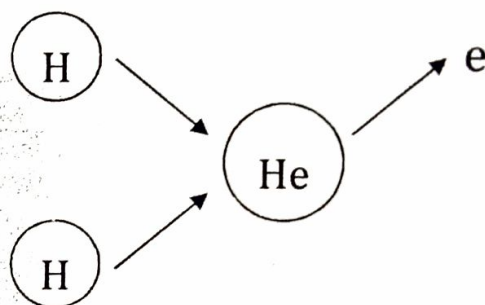


10.2 FISSION VS FUSION

Fission occurs when a large atom (element with high mass number) is hit with a high-energy particle (i.e. fast neutron) they break apart into two or more smaller elements. This produces energy and is the basis of nuclear power plants.



Fusion when smaller atoms combine to form a bigger atom. This produces more energy than fission but requires greater energy to induce. This is the basis of the energy production in the Sun.



10.3 HALF-LIFE

RATE OF DECAY and HALF-LIFE

- The mass of a radioactive material decreases as time passes.
- The amount of time it takes for a radioactive material to decay until half of the original amount is left is called **half-life**.
- The amount left in a radioactive material can be calculated as:

$$m_t = m_0 \left(\frac{1}{2}\right)^n \quad n = T/T_{\frac{1}{2}}$$

LESSON 11: ORGANIC CHEMISTRY

Organic Chemistry is the chemistry dealing with organic compounds

11.1 ORGANIC COMPOUNDS

These are compounds containing carbon and hydrogen or sometimes together with other elements such as oxygen, nitrogen and sulfur.

Hydrocarbons – organic compounds contains on carbon and hydrogen

Isomers – compounds having the same molecular formula but with a different structural formula

FUNCTIONAL GROUPS OF HYDROCARBONS

i. **Aliphatic Hydrocarbons** – molecules have their carbon atom skeleton arranged in straight or branched chains

- **Alkanes** – hydrocarbons containing single bonds only

Examples: $C_2H_6 \rightarrow$ ethane, $C_5H_{12} \rightarrow$ pentane

- **Alkenes** – hydrocarbons with at least one carbon to carbon double bond

Examples: $C_2H_4 \rightarrow$ ethene, $C_5H_{10} \rightarrow$ pentene

- **Alkynes** – hydrocarbons with one or more carbon to carbon triple bond

Examples: $C_2H_2 \rightarrow$ ethyne, $C_5H_8 \rightarrow$ pentyne

ii. **Aromatic Hydrocarbons** – organic substances with carbon atom skeleton arranged in a cyclic hexagonal arrangement such as benzene, C_6H_6 .

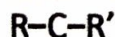
iii. **Alcohol** – contains the functional group – OH, with general formula of ROH. Aromatic alcohols are called **phenols**

Examples: CH_3CH_2OH

****Note:** R – stands for the carbon chain

X – stands for Halogens

iv. **Ether** – has the general formula



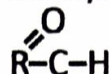
v. **Halide** – has the general formula



vi. **Amine** – has the general formula

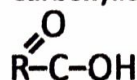


vii. **Aldehyde** – has the general formula

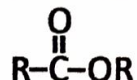


viii. **Ketone** – has the general formula $R-C(=O)-R$

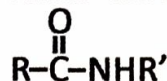
ix. **Carboxylic Acid** – has the general formula



x. **Ester** – has the general formula



xi. **Amide** – has the general formula



xii. **Acyl halide** – has the general formula $R-C(=O)-Cl$

xiii. **Acyl Anhydride** – has the general formula



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KEY

= Solid at room temperature
 = Liquid at room temperature
 = Gas at room temperature
 = Radioactive
 = Artificially Made

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